
Discovering Cooperative Pipelines: Autoresearch for Sequential Social Dilemmas

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Abstract

1 We study *two-level autoresearch* for cooperation: an outer-loop AI agent au-
2 tonomously redesigns the inner-loop pipeline of an LLM policy-synthesis sys-
3 tem for multi-agent Sequential Social Dilemmas (SSDs). A *researcher agent* $\mathcal{R}$
4 (Claude Opus 4.6, run as a coding agent) reads the inner-loop source code, edits
5 system prompts, feedback functions, helper libraries, and iteration logic, runs eval-
6 uations, and decides what to keep—following the *autoresearch* paradigm. Across
7 two games (Cleanup and Gathering), two policy-synthesizer LLMs (Claude Son-
8 net 4.6, Gemini 3.1 Pro), and two welfare objectives (utilitarian efficiency and
9 Rawlsian maximin), the researcher reliably exceeds hand-designed baselines,
10 sharply tightens run-to-run variance, and outperforms prompt-only optimization.
11 The discovered pipelines are objective-dependent: under maximin the researcher
12 independently rediscovers *duty rotation* as a fairness mechanism, supporting an
13 *information-design* reading in which the researcher chooses what to reveal to
14 the boundedly rational synthesizer. Code at [https://github.com/anon590/](https://github.com/anon590/llm-policies-social-dilemmas)
15 [llm-policies-social-dilemmas](https://github.com/anon590/llm-policies-social-dilemmas).

16 1 Introduction

17 Sequential Social Dilemmas (SSDs) [1] are the multi-agent analogue of the prisoner’s dilemma
18 in temporally rich Markov games: individually rational play leads to collectively suboptimal out-
19 comes through pollution, over-harvesting, or open conflict. Standard multi-agent reinforcement
20 learning (MARL) struggles in this regime due to credit assignment, non-stationarity, and large
21 joint action spaces [2]. A complementary approach, recently introduced by Gallego [3], replaces
22 parameter-space optimization with *algorithm-space synthesis*: a frozen LLM writes a Python policy
23 function, evaluates it in self-play, and iteratively refines it from performance feedback. A single
24 generation step can produce coordination logic—territory partitioning, role assignment, conditional
25 cooperation—that would take millions of RL episodes to discover.

26 This shifts where the design problem lives, rather than removing it. The inner-loop pipeline that
27 drives the synthesizer has many free parameters: which system prompt, which feedback variables,
28 which helper functions, how many refinement steps. Each materially affects the resulting policies,
29 and prior work tuned them by hand. A natural question follows: *can an AI agent design the pipeline?*

30 We answer affirmatively with a two-level autoresearch framework. An *outer-loop researcher agent*
31 (Claude Opus 4.6, run as a coding agent via the Claude Code CLI) edits the source files of an *inner-*
32 *loop policy synthesizer* (Gemini 3.1 Pro or Claude Sonnet 4.6), runs evaluations on held-out seeds,
33 and keeps modifications that improve a fixed welfare objective Φ . The outer agent operates on an
34 ordinary git repository—reading code, writing diffs, running shell commands—without task-specific
35 scaffolding, mirroring the autoresearch paradigm of Karpathy [4] for single-GPU LLM pretraining.
36 Although the inner-loop SSDs are gridworld benchmarks rather than physical systems, the outer-

loop discovery process itself runs under conditions a deployed discovery agent faces: noisy multi-seed evaluations, stochastic code generation, an LLM-evaluation budget that bounds how often Φ can be queried, and a heterogeneous Python repository the agent must navigate end-to-end on its own.

Our contributions are:

- A general two-level framework that delegates the design of an LLM synthesis pipeline to a coding agent operating on a real software repository (Section 3).
- The first instantiation of the autoresearch paradigm in a multi-agent decision-making domain, with 12 autonomous runs across two SSDs (Cleanup and Gathering), two policy LLMs, and two welfare objectives (utilitarian efficiency U and Rawlsian maximin $\min_i R_i$). The researcher reliably exceeds hand-designed baselines and dominates prompt-only optimization (GEPA) [5] (Section 4).
- A mechanism-design interpretation supported by the qualitatively different pipelines the agent produces under different welfare objectives—including the autonomous discovery of *time-based duty rotation* as a fairness mechanism, never found under efficiency optimization.

2 Background

We build on the iterative LLM policy synthesis framework of Gallego [3], which serves as the (frozen) inner loop of our two-level system. This section recalls the SSD formalism, the social outcome metrics, and the base synthesis loop.

2.1 Sequential Social Dilemmas

A Sequential Social Dilemma is a partially observable Markov game $\mathcal{G} = \langle N, \mathcal{S}, \{\mathcal{A}_i\}_{i=1}^N, T, \{R_i\}_{i=1}^N, H \rangle$ with N agents, state space $\mathcal{S}$ (the gridworld configuration), per-agent action spaces $\mathcal{A}_i$, transition function T , reward functions R_i , and episode horizon H [1]. Beyond the dilemma’s matrix-game structure, SSDs add temporal richness: agents must learn *when* and *where* to cooperate, not just *whether* to.

We study two canonical SSDs that capture complementary dilemma types.

Cleanup [6] is a *public goods provision* game ($N=10$). The map has two regions: a river that accumulates waste, and an orchard where apples grow. Apples regrow only when river pollution is below threshold. Each agent can fire a cleaning beam (cost -1) that removes waste, collect apples ($+1$ each), or fire a tagging beam (cost -1 , inflicting -50 on the target and removing it for 25 steps). The dilemma: cleaning is costly but its benefits are public, so purely self-interested agents free-ride.

Gathering [1, 7] is a *common pool resource* game ($N=4$). Agents collect shared apples on a fixed respawn timer and may fire tagging beams to temporarily remove rivals. The dilemma: agents can coexist and share resources, or aggress to monopolize them; aggression wastes time and reduces total welfare.

Both games use 8–9 discrete actions (movement, rotation, beam, stand, optionally clean) and episodes of $H=1000$ steps. The two dilemmas differ in cost structure—asymmetric provision (cleaners pay, all benefit) vs. symmetric restraint (every agent faces the same temptation)—a distinction that drives our key findings.

75 **Social metrics.** Following Perolat et al. [7], let $R_i = \sum_{t=0}^{H-1} r_i^t$ denote agent i 's episode return.
 76 We evaluate four social outcomes:

$$U = \frac{1}{H} \sum_{i=1}^N R_i \quad (\text{efficiency}) \quad (1)$$

$$E = 1 - \frac{\sum_{i,j} |R_i - R_j|}{2N \sum_i R_i} \quad (\text{equality}) \quad (2)$$

$$S = \frac{1}{N} \sum_{i=1}^N \bar{t}_i \quad (\text{sustainability}) \quad (3)$$

$$P = \frac{1}{H} \sum_{t=0}^{H-1} |\{i : \text{active}_i^t\}| \quad (\text{peace}) \quad (4)$$

77 where $\bar{t}_i$ is the mean timestep at which agent i collects positive reward (higher $\Rightarrow$ resources preserved
 78 later) and active_i^t indicates that agent i is not tagged out at step t . We additionally consider the
 79 *maximin* (Rawlsian) welfare criterion $\min_i R_i$, which optimizes the worst-off agent's return and
 80 serves as the second objective in our experiments.

81 2.2 Iterative LLM Policy Synthesis

82 Let Π denote the space of *programmable policies*: deterministic functions $\pi : \mathcal{S} \times [N] \rightarrow \mathcal{A}$ ex-
 83 pressed as executable Python code. Each policy has access to the full environment state and a
 84 library of helpers (BFS pathfinding, beam targeting, coordinate transforms). This state access is a
 85 deliberate design choice: programmatic policies operate in algorithm space rather than the reactive
 86 observation-to-action space of neural policies, which lets a single LLM generation step encode rich
 87 coordination logic.

88 A frozen LLM $\mathcal{M}$ acts as the *policy synthesizer*. Given a system prompt p describing the environ-
 89 ment API and a feedback prompt q_k , it produces a new policy

$$\pi_{k+1} = \mathcal{M}(p, q(\pi_k, \mathcal{F}_k)),$$

90 where π_k is the previous policy (its source code) and $\mathcal{F}_k$ is the evaluation feedback. All N agents ex-
 91 ecute the same policy in homogeneous self-play, and evaluation over a set of random seeds S yields
 92 the mean per-agent return $\bar{r}_k$ and the social metrics vector $\mathbf{m}_k = (U_k, E_k, S_k, P_k)$. Each generated
 93 policy passes an AST-based safety check (blocking eval, file I/O, network access) followed by a
 94 short smoke test; failures trigger regeneration (up to R attempts) with the error message appended
 95 to the prompt.

96 **Feedback.** The hand-designed feedback used by Gallego [3] as our starting point packages the
 97 previous policy's source code together with all available evaluation signals:

$$\mathcal{F}_k = (\text{code}(\pi_k), \bar{r}_k, \mathbf{m}_k, \mathbf{d}), \quad (5)$$

98 where $\mathbf{d}$ contains natural-language definitions of each social metric. Exposing the full social-metrics
 99 vector (rather than the scalar reward alone) disambiguates distinct failure modes that share the same
 100 average return—e.g., under- vs. over-cleaning in Cleanup—and consistently improves cooperative
 101 outcomes. On Cleanup, this baseline feedback reaches $U=1.93/2.70$ (mean/max) with Gemini and
 102 $U=0.86/1.56$ with Sonnet across 4 runs each. The choice of feedback content is a single design
 103 decision within a much larger pipeline configuration space; our two-level framework (Section 3)
 104 opens the full space to automated search.

105 3 Two-Level Framework

106 We introduce a two-level system where a *researcher agent* $\mathcal{R}$ autonomously discovers configurations
 107 that optimize the output of an inner-loop system. While we instantiate this for multi-agent policy
 108 synthesis, the architecture is general: any pipeline where an LLM generates artifacts, evaluates them,
 109 and iterates can serve as the inner loop. The key insight is that the entire inner-loop codebase—not
 110 just the feedback level—is a designable artifact that an AI agent can search over. Figure 1 illustrates
 111 the architecture.

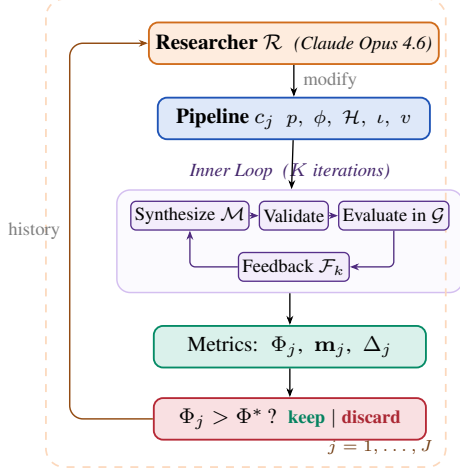


Figure 1: Two-level automated research framework (Algorithm 1).

Algorithm 1 Two-Level Automated Research

Require: Game $\mathcal{G}$, policy synthesizer $\mathcal{M}$, researcher $\mathcal{R}$, system prompt $p_{\mathcal{R}}$, initial config c_0 , outer iterations J , ground-truth objective Φ , held-out seeds S_{ho}

Ensure: Best configuration c^* , best policy π^*

```

1:  $\pi_0^* \leftarrow \text{INNERLOOP}(\mathcal{M}, \mathcal{G}, c_0)$ 
2:  $\Phi_0 \leftarrow \text{EVAL}(\pi_0^*; \mathcal{G}, S_{\text{ho}})$ 
3:  $\text{history} \leftarrow \{(c_0, \Phi_0, \mathbf{m}_0, \emptyset)\}$ 
4: for  $j = 1, \dots, J$  do
5:    $c_j \leftarrow \mathcal{R}(p_{\mathcal{R}}, \text{CODE}(c_{j-1}), \text{history})$ 
6:   if  $\neg \text{VALIDATECONFIG}(c_j)$  then  $\text{retry} \leq R$ 
7:      $\pi_j^* \leftarrow \text{INNERLOOP}(\mathcal{M}, \mathcal{G}, c_j)$ 
8:      $\Phi_j \leftarrow \text{EVAL}(\pi_j^*; \mathcal{G}, S_{\text{ho}})$ 
9:      $\Delta_j \leftarrow \text{DIFF}(c_{j-1}, c_j)$  // code diff
10:     $\text{history} \leftarrow \text{history} \cup \{(c_j, \Phi_j, \mathbf{m}_j, \Delta_j)\}$ 
11:  end for
12:  $j^* \leftarrow \arg \max_j \Phi_j$ 
13: return  $c_{j^*}, \pi_{j^*}^*$ 

```

Table 1: Configuration components modifiable by the researcher. The environment simulator, ground-truth evaluation, and policy LLM weights are frozen.

	Scope	Examples
p	System prompt	Hints, worked examples
ϕ	Feedback fn	Metric subset, framing, derived metrics, trends
$\mathcal{H}$	Helper library	Pathfinding, zones
ι	Iteration logic	K , eval seeds, best-of- n
v	Validation	Safety checks, smoke tests

112 3.1 Configuration Space

113 Let $\mathcal{C}$ denote the space of *pipeline configurations*. Each configuration $c \in \mathcal{C}$ specifies the full inner-
 114 loop setup:

$$c = (p, \phi, \mathcal{H}, \iota, v) \quad (6)$$

115 where p is the system prompt, ϕ is the feedback construction function (which content to include, how
 116 to frame it, whether to inject adaptive hints), $\mathcal{H}$ is the helper function library, ι specifies the iteration
 117 logic (number of iterations K , sampling strategy, temperature schedule), and v is the validation
 118 pipeline. Table 1 provides concrete examples.

119 The hand-designed feedback of [3] corresponds to a single fixed instantiation of ϕ . Our framework
 120 opens the full configuration space to automated search.

121 3.2 Inner Loop (Policy Synthesis)

122 Given a configuration c , the inner loop executes K iterations of LLM policy synthesis:

$$\pi_c^* = \text{INNERLOOP}(\mathcal{M}, \mathcal{G}, c) \quad (7)$$

123 Each iteration k proceeds in four stages, following [3]:

- 124 1. **Synthesize.** The policy LLM $\mathcal{M}$ receives the system prompt p , the previous policy’s source code
 125 π_{k-1} , and feedback $\mathcal{F}_{k-1}$ constructed by ϕ . It generates a new Python policy function π_k that
 126 has access to full environment state and the helper library $\mathcal{H}$.
- 127 2. **Validate.** The generated code undergoes AST-based safety checks (blocking dangerous oper-
 128 ations such as file I/O and network access) followed by a short smoke test. Failures trigger
 129 re-generation (up to R retries), with the error message appended to the prompt.
- 130 3. **Evaluate.** All N agents execute the same policy π_k in self-play over $|S|$ random seeds. The eval-
 131 uation yields the mean per-agent reward $\bar{r}_k$ and the social metrics vector $\mathbf{m}_k = (U_k, E_k, S_k, P_k)$.

132 **4. Feedback.** The feedback function ϕ constructs the prompt for the next iteration from
 133 $(\pi_k, \bar{r}_k, \mathbf{m}_k)$, packaging the previous policy’s source code together with the scalar reward, the
 134 social metrics vector, their natural-language definitions, and any adaptive diagnostics that ϕ in-
 135 jects.

136 The inner loop output is evaluated on held-out seeds via a fixed ground-truth objective:

$$\Phi(c) = \text{EVAL}(\pi_c^*; \mathcal{G}, S_{\text{held-out}}) \quad (8)$$

137 where Φ maps from social outcomes to a scalar. We consider two objectives:

$$\Phi_U = U \quad (\text{utilitarian efficiency}) \quad (9)$$

$$\Phi_{\min} = \min_i R_i \quad (\text{Rawlsian maximin}) \quad (10)$$

138 3.3 Outer Loop (Automated Research)

139 The researcher agent $\mathcal{R}$ iteratively modifies the pipeline configuration. Following the autoresearch
 140 paradigm [4], $\mathcal{R}$ operates on the inner-loop codebase as a modifiable artifact, proposing changes,
 141 observing outcomes, and refining. The procedure is formalized in Algorithm 1.

142 At each outer iteration j , the researcher $\mathcal{R}$ receives:

- 143 • The **full source code** of the current configuration c_{j-1} (prompts, feedback construction, helpers,
 144 iteration logic).
- 145 • The **experiment history**: for each prior iteration, the code diff Δ_i , ground-truth score Φ_i , and
 146 social metrics vector $\mathbf{m}_i$.
- 147 • The **environment source code** (read-only), enabling the researcher to reason about game mechan-
 148 ics.

149 The researcher proposes a new configuration c_j by generating code modifications. Concretely, $\mathcal{R}$
 150 is a coding agent (Claude Code CLI) that operates on a real software repository: it reads and edits
 151 Python source files, runs shell commands, inspects evaluation outputs, and commits changes to
 152 a dedicated git branch—the same workflow a human researcher would follow. This mirrors the
 153 autoresearch setup, where an AI coding agent modifies `train.py` and observes validation loss;
 154 here, the researcher modifies the inner-loop codebase and observes social welfare.

155 3.4 Connection to Automated Mechanism Design

156 The two-level structure admits a mechanism design interpretation. The researcher $\mathcal{R}$ acts as a *mech-*
 157 *anism designer*: it controls the information structure (what metrics to reveal, how to frame them),
 158 the action space (what helper functions are available), and the incentive structure (how feedback is
 159 presented) under which the policy synthesizer $\mathcal{M}$ operates. The synthesizer acts as the *agent* within
 160 the designed mechanism.

161 This connects to the automated mechanism design literature [8], where a principal designs rules to
 162 induce desired behavior from self-interested agents. In our setting:

- 163 • The **principal** is the researcher $\mathcal{R}$, optimizing the ground-truth objective Φ .
- 164 • The **agent** is the synthesizer $\mathcal{M}$, optimizing per-agent reward as instructed.
- 165 • The **mechanism** is the configuration c : prompts, feedback, helpers, iteration logic.
- 166 • The **outcome** is the social welfare of the resulting multi-agent policy π_c^* .

167 A key distinction from classical mechanism design: $\mathcal{M}$ is not *strategically* self-interested—it fol-
 168 lows instructions—but it has *bounded rationality* in the sense that its ability to synthesize effective
 169 policies depends on the information and tools provided. The researcher’s task is thus closer to *infor-*
 170 *mation design* [9]: choosing what to reveal to help $\mathcal{M}$ navigate the cooperation–defection tension.
 171 Our experiments (Section 4) show that the researcher designs qualitatively different information
 172 structures depending on the welfare objective Φ , supporting this interpretation empirically.

4 Experiments

4.1 Setup

We conduct 12 autonomous researcher runs across a factorial design. The researcher agent $\mathcal{R}$ is Claude Opus 4.6, invoked via the Claude Code CLI as a coding agent. Each run operates on a dedicated git branch of a real Python codebase: $\mathcal{R}$ edits source files in `pipeline/`, executes evaluation scripts, reads metric outputs, and iterates—without human intervention.

Design. For Cleanup: 2 policy LLMs $\times$ 2 objectives $\times$ 2 replications = 8 runs. For Gathering: 2 policy LLMs $\times$ 1 objective $\times$ 2 replications = 4 runs. Maximin runs are unnecessary for Gathering because efficiency optimization alone achieves equality $E > 0.94$ (the game’s symmetric cost structure makes fairness a free byproduct of coordination).

Models. Policy synthesizer $\mathcal{M}$: Gemini 3.1 Pro (Google) or Claude Sonnet 4.6 (Anthropic). Both use extended thinking. These are the same models evaluated in [3]. We additionally test with Gemma 4 26B-A4B-IT (Google), a smaller open-weight model, to probe the framework’s behavior when the policy synthesizer has substantially lower capability (Appendix C).

Baselines. The hand-designed feedback configuration from [3] serves as the initial pipeline c_0 for all runs. On Cleanup ($N=10$), this baseline achieves $U=1.93/2.70$ (mean/max) with Gemini and $U=0.86/1.56$ with Sonnet across 4 runs each. We additionally compare against GEPA [5], an automated prompt optimization method that iteratively refines the system prompt via LLM reflection. GEPA optimizes only the system prompt p , whereas our researcher modifies the full pipeline $(p, \phi, \mathcal{H}, \iota, v)$. We give GEPA a matched compute budget: the same number of optimization steps as outer iterations used by our method (each GEPA step costs ~ 3 policy evaluations, comparable to our inner loop’s $K+1=4$).

Inner loop parameters. $K=3$ synthesis iterations (or $K=2$ if the researcher elects), evaluated over $|S|=5\text{--}12$ random seeds (researcher-configurable). The researcher ran $J=3\text{--}17$ outer iterations per run (mean: 8.3), with a keep rate of 18%–100% (mean: 63%).

4.2 Results

Table 2 presents the main results, comparing autoresearch against the hand-designed baseline of [3] and GEPA prompt optimization [5].

Finding 1: The researcher reliably improves over hand-designed baselines and outperforms prompt-only optimization. All 12 runs show substantial improvement regardless of starting point (Figure 2). In Cleanup, the researcher exceeds the unmodified pipeline baseline by 66% with Gemini ($U: 1.93 \rightarrow 3.20$) and 263% with Sonnet ($U: 0.86 \rightarrow 3.12$). Strikingly, the Sonnet improvement nearly closes the gap with Gemini ($U=3.12$ vs. 3.20), despite a $2\times$ baseline disadvantage (0.86 vs. 1.93). Final efficiency values cluster tightly around $U \approx 3.1\text{--}3.2$ despite baselines varying from 0.29 to 2.70 across individual runs, suggesting the researcher reliably discovers the performance ceiling of each policy LLM. Beyond improving the mean, the researcher yields remarkably consistent results: under efficiency optimization, Gemini’s runs cluster within $U=3.20\text{--}3.25$ (gap 0.05) despite baselines spanning 1.15–2.70, and Sonnet’s within 3.12–3.14 (gap 0.02) from baselines of 0.38–1.56. In Gathering, all four runs converge to $U=2.47\text{--}2.52$ from baselines ranging 0.03–2.42. The researcher discovers pipeline modifications—structured helpers (App. D.1), strategic hints (App. D.2), tailored feedback (App. D.3)—that constrain the policy LLM toward consistently good solutions.

Compared to GEPA (Table 2), autoresearch achieves $2.4\times$ higher efficiency under Φ_U in Cleanup ($U=3.20$ vs. 1.34), $2\times$ higher maximin under $\Phi_{\min}$ (290 vs. 143), and 20% higher efficiency in Gathering ($U=2.49$ vs. 2.08). Critically, GEPA shows far wider spread across runs on the harder fairness objective ($\Phi_{\min}$ maximin in Cleanup: 245/41 vs. autoresearch’s 296/284). The gap widens further with Sonnet: on Cleanup Φ_U , GEPA stalls at $U=1.04$ ($3\times$ below autoresearch’s 3.12), and on $\Phi_{\min}$ its two runs split between a modestly positive solution ($\min_i R_i=6$, $U=1.60$) and a pathological “everyone cleans, nobody eats” regime (perfect equality $E=1.00$ but negative efficiency $U=-2.87$).

Table 2: Results. Mean / max across 2 runs per condition (4 for Cleanup baselines). Baseline: unmodified pipeline from [3]. GEPA: automated prompt optimization [5] with matched compute budget.

Policy LLM	Target	U	E	$\min_i R_i$
<i>Cleanup (N=10) — Baseline</i>				
Gemini 3.1 Pro	—	1.93/2.70	0.17/0.62	−159/−84
Sonnet 4.6	—	0.86/1.56	−0.02/0.25	−151/−59
<i>Cleanup (N=10) — GEPA [5]</i>				
Gemini 3.1 Pro	Φ_U	1.34/1.37	−0.10/−0.05	−149/−126
	$\Phi_{\min}$	1.76/2.76	0.90/0.96	143/245
Sonnet 4.6	Φ_U	1.04/1.20	−0.59/−0.21	−164/−126
	$\Phi_{\min}$	−0.63/1.60	0.77/1.00	−147/6
<i>Cleanup (N=10) — Autoresearch</i>				
Gemini 3.1 Pro	Φ_U	3.20 /3.25	0.55/0.61	−211/−182
	$\Phi_{\min}$	3.16/3.19	0.98 /0.98	290 /296
Sonnet 4.6	Φ_U	3.12 /3.14	0.66/0.70	−196/−146
	$\Phi_{\min}$	2.57/2.93	0.91 /0.97	179 /200
<i>Gathering (N=4) — Baseline</i>				
Gemini 3.1 Pro	—	2.04/2.42	0.90/0.98	412/571
Sonnet 4.6	—	0.03/0.03	0.54/0.54	0/0
<i>Gathering (N=4) — GEPA [5]</i>				
Gemini 3.1 Pro	Φ_U	2.08/2.35	0.94/0.96	436/518
Sonnet 4.6	Φ_U	1.20/1.23	0.63/0.71	86/123
<i>Gathering (N=4) — Autoresearch</i>				
Gemini 3.1 Pro	Φ_U	2.49/2.51	0.98/0.98	582/582
Sonnet 4.6	Φ_U	2.52/2.52	0.96/0.98	576/597

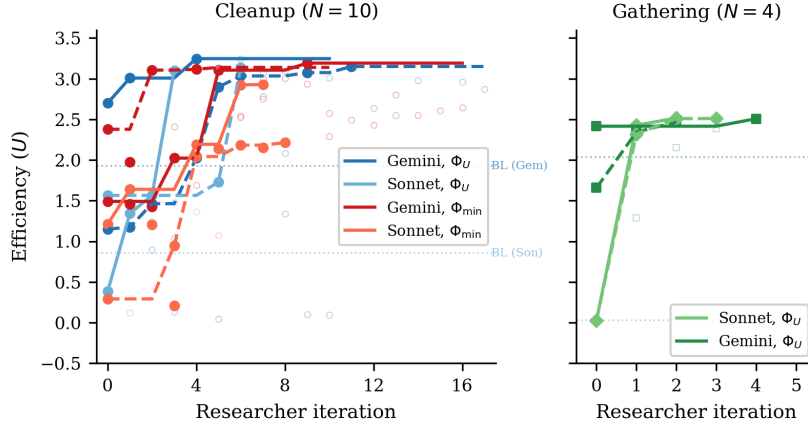


Figure 2: Efficiency (U) across researcher iterations for all 12 runs. *Left*: Cleanup ($N=10$) with 8 runs across 2 LLMs $\times$ 2 objectives (solid lines = kept experiments, open circles = discarded). Dashed horizontal line: hand-designed baseline from [3]. All runs converge to $U \approx 3.1$ – 3.2 despite diverse starting points. *Right*: Gathering ($N=4$) with 4 runs (2 per LLM). All converge to $U \approx 2.5$ within 2–4 iterations despite baselines ranging from 0.03 to 2.42.

222 and $\min_i R_i = -299$, while autoresearch with the same model reaches $\min_i R_i = 179/200$ reliably.
223 This highlights the advantage of modifying the full pipeline—helpers, feedback, and iteration logic
224 provide structured scaffolding that reduces run-to-run variance, whereas prompt-only optimization
225 cannot address the underlying stochasticity of code generation.

226 **Finding 2: No efficiency–fairness tradeoff in Cleanup (Gemini).** Maximin-optimized Gem-
227 ini pipelines sacrifice only 1% efficiency (U : 3.16 vs. 3.20) while achieving near-perfect equal-
228 ity (E : 0.98 vs. 0.55) and transforming maximin from deeply negative baselines (−99 to −84) to

229 $\min_i R_i = 290$. The researcher discovers *fair duty rotation* (Listing 6; primed by the prompt rewrite
 230 of Listing 3)—time-based cycling using `agent_id` and `env._step_count`—simultaneously im-
 231 proving worst-off welfare *and* collective output. Because cleaning is a public good, distributing the
 232 cleaning cost fairly ensures enough cleaners to sustain apple production.

233 For Sonnet, there is a moderate tradeoff: efficiency drops from 3.12 to 2.57 under maximin opti-
 234 mization. The gap reflects Sonnet’s harder time implementing complex coordination mechanisms
 235 (role rotation, zone assignment) from strategic hints alone.

236 **Finding 3: Game structure determines whether fairness requires explicit optimization.** In
 237 Cleanup, where cleaning costs are borne asymmetrically (cleaners pay -1 , free-riders collect ap-
 238 ples), baseline equality ranges from $E=0.04$ to 0.62 , and maximin optimization is required to reach
 239 $E > 0.9$. In Gathering, where all agents face a symmetric landscape, efficiency optimization alone
 240 achieves $E > 0.94$ across all 4 runs—no separate maximin runs are needed.

241 This generalizes: in *provision* dilemmas with asymmetric costs, fairness requires designed mecha-
 242 nisms (role rotation, duty sharing); in *restraint* dilemmas with symmetric costs, fairness emerges as
 243 a free byproduct of efficient coordination. The researcher independently discovers this—it creates
 244 role differentiation pipelines only for Cleanup, and pure spatial-coordination pipelines for Gather-
 245 ing.

246 **Finding 4: Convergent discovery of qualitatively different strategies per objective.** Despite
 247 fully independent runs, the researcher converges on the same core strategies within each condition
 248 (Table 3, Appendix A). In Cleanup, waste-counting helpers and spatial zone partitioning appear
 249 across all 8 runs. The key qualitative difference is *role rotation*, appearing in 4/4 maximin runs
 250 (Listings 6, 7) but 0/4 efficiency runs (Listing 5). Under efficiency optimization, the researcher
 251 assigns static cleaning roles (some agents always clean), achieving high collective output at the cost
 252 of equality. Under maximin optimization, it discovers rotation as a fairness mechanism.

253 In Gathering, the researcher discovers BFS-Voronoi territory partitioning and respawn-timer aware-
 254 ness (Listings 2, 8), with no role differentiation—optimization is purely spatial (who collects which
 255 apples) and temporal (respawn-aware positioning).

256 **Common failure modes.** Several patterns led to discarded experiments across all runs: (1) *over-*
 257 *prescription*—adding too many strategic hints confuses the policy LLM, causing generation failures;
 258 (2) *iteration regression*— $K=4$ or $K=5$ inner iterations often cause catastrophic regression as the
 259 LLM “over-refines” a working policy; (3) *feedback overload*—verbose per-agent diagnostics cause
 260 over-correction rather than gradual improvement; (4) *variance exposure*—identical configurations
 261 can yield $U=3.2$ then $U=0.09$ on different seeds, requiring multi-seed evaluation for stable signals
 262 (per-run $K, |S|$ in Table 6).

263 5 Related Work

264 **Sequential social dilemmas.** SSDs were introduced by Leibo et al. [1] (Gathering) and extended
 265 to public-goods settings by Hughes et al. [6] (Cleanup); Perolat et al. [7] formalized the social
 266 outcome metrics (U, E, S, P) used here. Subsequent work studies inequity aversion, intrinsic moti-
 267 vation, and reputational mechanisms for promoting cooperation in MARL agents. We complement
 268 this line by automating the search for cooperative *programs* rather than evolving neural policies,
 269 and by treating the welfare objective Φ as a designable parameter that the outer agent optimizes
 270 for—obtaining qualitatively different cooperative behaviors (static role assignment vs. duty rotation)
 271 under different objectives without changing the environment.

272 **LLMs for policy and program synthesis.** FunSearch [10] evolves programs for combinato-
 273 rial discovery; Eureka [11] synthesizes reward functions from environment source code; Voy-
 274 ager [12] and Code as Policies [13] generate executable skill code for single-agent embodied control;
 275 ReEvo [14] evolves heuristics with reflective feedback. These works target single-agent settings or
 276 non-strategic optimization. Gallego [3], on which our inner loop is based, extended LLM program
 277 synthesis to the multi-agent SSD setting, where one program must coordinate N self-play copies.
 278 Our contribution is one level above: rather than tuning the synthesizer for one task, we let an outer
 279 agent rewrite the synthesis pipeline itself.

LLM reflection and prompt optimization. Reflexion [15], Self-Refine [16], OPRO [17], and GEPA [5] demonstrate that structured verbal feedback loops improve LLM outputs; ERL [18] internalizes such reflection via self-distillation. These methods optimize the prompt or the model’s reasoning trajectory in isolation. Our framework instead modifies the entire surrounding pipeline—system prompt, feedback construction, helper library, iteration logic, validation—making prompt optimization (cf. GEPA) one component of a strictly larger search space. The empirical gap is large: Section 4 shows full-pipeline autoresearch achieving $3\times$ higher efficiency in Cleanup and 37% higher in Gathering than GEPA at matched compute, with run-to-run spread an order of magnitude tighter on the harder fairness objective.

Automated AI research. A small but growing body of work delegates the design of ML pipelines to LLM coding agents. Karpathy’s autoresearch [4] runs a coding agent that modifies `train.py` for nanoGPT pretraining and is rewarded for validation loss. Our system shares this autoresearch architecture—coding agent + frozen evaluation harness + diff-based history—but targets a different inner loop (multi-agent policy synthesis instead of single-model pretraining) and a different objective (multi-agent social welfare instead of validation bits-per-byte). To our knowledge this is the first instantiation of the autoresearch paradigm in a multi-agent decision-making domain.

Automated mechanism and information design. Classical automated mechanism design [8] computes optimal allocation rules for self-interested agents under explicit incentive constraints, while information design [9] studies how a principal commits to a signaling policy that shapes a receiver’s behavior. Our outer agent solves a related but distinct problem: $\mathcal{M}$ is not strategically deceptive (it follows instructions) but is *boundedly rational*, so the researcher must decide what information, helpers, and structure to expose so that $\mathcal{M}$ writes policies achieving the principal’s welfare goal. Section 4 shows that the pipelines $\mathcal{R}$ produces are genuinely a function of the welfare objective, supporting this information-design framing empirically.

6 Discussion and Conclusion

We have presented a two-level framework where an AI coding agent autonomously discovers pipeline configurations that improve the output of an inner-loop LLM system. Applied to multi-agent policy synthesis in social dilemmas, the researcher agent reliably exceeds hand-designed baselines across 12 independent runs, and converges on qualitatively similar strategies within each game-objective condition. The framework requires no domain-specific scaffolding: the agent operates on a standard software repository using the same tools (file editing, shell commands, git) available to a human researcher.

Mechanism design in action. Our results empirically support the mechanism design interpretation of Section 3.4. The researcher acts as an information designer: under Φ_U , it reveals efficiency-oriented information (waste counts, zone assignments) that guides $\mathcal{M}$ toward productive but unequal role allocation (Listing 5). Under $\Phi_{\min}$, it additionally reveals fairness-oriented structure—rotation schedules and per-agent equity feedback (Listings 3, 4)—guiding $\mathcal{M}$ toward egalitarian coordination. The striking absence of a fairness-efficiency tradeoff with Gemini suggests that in public goods provision, the mechanism design problem has a cooperative optimum where both objectives align—the researcher discovers this independently.

Two types of dilemma. The Cleanup–Gathering contrast reveals a structural insight about social dilemmas: *provision* dilemmas (asymmetric costs, one agent’s contribution benefits all) require explicit fairness mechanisms, while *restraint* dilemmas (symmetric costs, each agent faces the same temptation) achieve fairness naturally when coordination improves. This distinction, well-known in the common-pool resource literature, is rediscovered here by an autonomous AI agent—suggesting it is a robust structural feature of these environments.

Human oversight and audit. Our system is fully autonomous by design, but its architecture offers natural affordances for human-in-the-loop oversight. Because the researcher $\mathcal{R}$ operates on a standard git repository—committing each pipeline modification to a dedicated branch—a domain expert can audit the full history of code diffs Δ_j alongside their metric outcomes $(\Phi_j, \mathbf{m}_j)$, exactly as they would review a colleague’s pull requests. The keep/discard decision (Algorithm 1, line 11)

is a lightweight intervention point: a human could override the accept threshold, veto modifications that introduce undesirable mechanisms (e.g., exploitative role assignments), or steer the researcher toward under-explored regions of the configuration space by editing the experiment history. More broadly, the separation between the researcher $\mathcal{R}$ and the frozen evaluation Φ creates a *delegation boundary*: the human defines *what* to optimize (the welfare objective), while the agent decides *how*. This mirrors the expert-in-the-loop design patterns identified in deployment-oriented work on AI agents for discovery, where trust calibration depends on the legibility of the agent’s decision trail rather than on real-time supervision.

Limitations. Our experiments use only 2 replications per condition, yielding high uncertainty on some estimates (e.g., Sonnet $\Phi_{\min}$: $U=2.57/2.93$). The two SSDs are small gridworld environments; scaling to larger, more complex games remains open. The researcher runs are not fully controlled—the number of outer iterations varies from 3 to 17 (mean 8.3)—making direct comparison across conditions approximate. The GEPA comparison uses a single run per condition; additional replications would strengthen confidence in the comparison.

Future work. Five directions emerge: (1) *new domains*—apply the two-level framework to other LLM-driven pipelines (code optimization, scientific experiment design, infrastructure tuning), where the same architecture applies with a different inner loop and objective Φ ; (2) *ablation*—freeze all configuration components except one to isolate the relative importance of prompt, feedback, helpers, and iteration logic; (3) *adversarial objectives*—set $\Phi = R_0$ (agent 0’s reward only) to test whether the researcher discovers exploitative pipeline configurations, connecting to the reward hacking analysis of [3]; (4) *cross-game transfer*—run the best Cleanup configuration on Gathering and vice versa, testing game-generalizability; (5) *heterogeneous policies*—extend to settings where agents run different code, enabling richer role specialization.

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A Additional Results

Table 3: Strategies discovered by the researcher in Cleanup across 4 efficiency-optimized and 4 maximin-optimized independent runs.

Strategy	Φ_U (4 runs)	$\Phi_{\min}$ (4 runs)
Waste-counting helpers	4/4	4/4
Zone/lane partitioning	3/4	4/4
Anti-regression feedback	3/4	2/4
Worked policy examples	2/4	3/4
Cleaning cost economics	2/4	3/4
Time-based role rotation	0/4	4/4

B Compute Requirements

Table 4 reports wall-clock time for all 12 autonomous runs and the 6 GEPA comparison runs. *Inner loop* measures total time spent on policy LLM generation and environment simulation across all evaluations; *total* (estimated from run-directory timestamps) additionally includes the researcher agent’s analysis, code editing, and decision-making between evaluations.

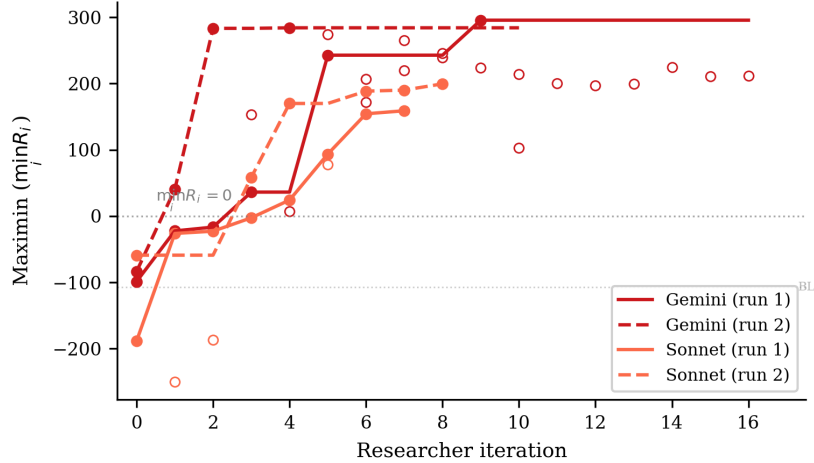


Figure 3: Maximin ($\min_i R_i$) across researcher iterations for the 4 Cleanup maximin-optimized runs. All runs transform deeply negative baselines (worst-off agents losing reward) into substantially positive values. Gemini reaches ~ 290 while Sonnet reaches ~ 160 – 200 . The dashed line marks $\min_i R_i = 0$ (no agent loses reward overall).

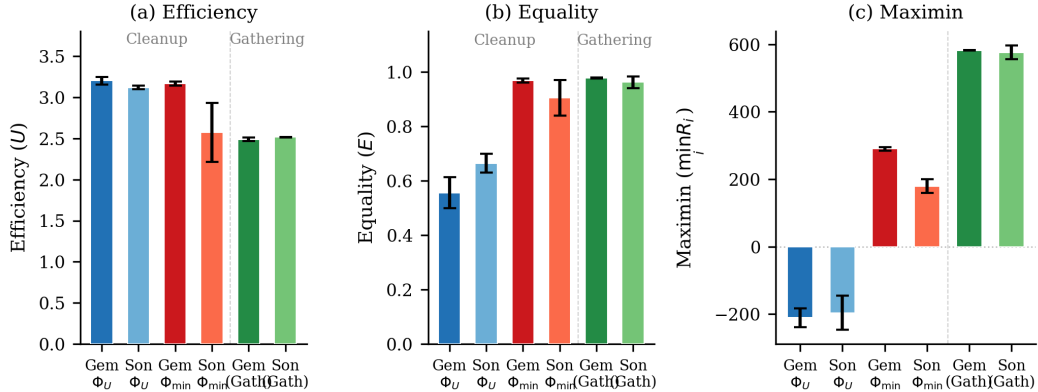


Figure 4: Final metrics across all conditions. (a) Efficiency: all conditions converge to $U \approx 2.5$ – 3.2 . (b) Equality: maximin-optimized Cleanup runs achieve $E \approx 1.0$, while efficiency-optimized runs show $E \approx 0.5$ – 0.7 ; Gathering achieves high equality regardless. (c) Maximin: the sharpest contrast—efficiency optimization leaves the worst-off agent deeply negative ($\min_i R_i \approx -200$), while maximin optimization transforms it to $+290$ (Gemini) or $+179$ (Sonnet). Gathering achieves high maximin (~ 580) under efficiency optimization alone. Error bars show s.d. across runs.

Cost breakdown. Each inner loop evaluation involves $K=2$ – 3 policy LLM generation calls (each $\sim 2k$ input tokens, $\sim 1.5k$ output tokens) followed by multi-seed simulation (5–12 seeds, ~ 15 – $30s$ total). Policy LLM generation dominates inner loop time (86–97%), with Sonnet evaluations taking $\sim 2\times$ longer than Gemini due to extended thinking. The researcher agent (Claude Opus 4.6, running via Claude Code CLI) accounts for 41% of total wall-clock time (25.6h of the 62.2h total across all 12 runs), spent reading results, editing pipeline source files, and planning modifications. In monetary terms, the researcher agent is the dominant cost: each outer iteration consumes ~ 50 – $100k$ context tokens in the Opus session, whereas each inner loop evaluation uses only ~ 5 – $10k$ policy LLM tokens. The 6 GEPA comparison runs required 5.0h of compute with no researcher agent, operating at lower cost per run but achieving substantially lower performance (Table 2).

Table 4: Wall-clock time per autonomous run. Evals: total inner loop evaluations including initial baseline. Per eval: mean wall-clock per single evaluation. Total: end-to-end including researcher overhead.

Policy LLM	Φ	Evals	Inner loop (h)	Per eval (min)	Total (h)
<i>Cleanup</i> ($N=10$)					
Gemini	Φ_U	11	2.6	14	3.7
Gemini	Φ_U	18	4.1	14	6.4
Sonnet	Φ_U	4	2.1	32	6.1
Sonnet	Φ_U	7	5.0	43	6.1
Gemini	$\Phi_{\min}$	17	3.0	11	4.3
Gemini	$\Phi_{\min}$	11	3.6	20	5.0
Sonnet	$\Phi_{\min}$	8	4.5	33	8.8
Sonnet	$\Phi_{\min}$	9	5.3	36	13.2
<i>Gathering</i> ($N=4$)					
Sonnet	Φ_U	3	1.7	33	2.2
Gemini	Φ_U	5	1.4	17	1.9
Gemini	Φ_U	3	0.9	19	1.1
Sonnet	Φ_U	4	2.3	35	3.4
All 12 runs		100	36.6	22	62.2
GEPA (6 runs)		6	5.0	50	—

424 C Smaller Open-Weight Model: Gemma 4 26B

425 We test the framework with Gemma 4 26B-A4B-IT (Google), a 26B-parameter open-weight model
426 substantially smaller than the frontier models in the main experiments. We run three experiments—
427 two on Cleanup ($N=10$) optimizing efficiency and maximin respectively, and one on Gathering
428 ($N=4$) optimizing efficiency—all with Opus 4.6 as the researcher $\mathcal{R}$.

429 **Setup.** In all three experiments, Gemma starts from complete failure: the baseline pipeline pro-
430 duces $U=-10.0$ on Cleanup (agents CLEAN-spam without collecting apples) and $U=0.0$ on Gath-
431 ering (broken BFS calls), compared to $U=1.93/0.86$ (Gemini/Sonnet on Cleanup) and $U=2.04/0.03$
432 (Gemini/Sonnet on Gathering). The researcher ran $J=10-18$ outer iterations per condition.

433 **Results.** Table 5 presents the best configurations discovered. Under efficiency optimization, the
434 researcher achieves $U=0.87$ —a dramatic recovery from the broken baseline but far below fron-
435 tier models ($U \approx 3.2$). The researcher compensates for the model’s limited capability by reducing
436 inner-loop iterations to $K=1$ (avoiding the regression that plagued $K \geq 2$ runs) and providing highly
437 structured worked examples with explicit cleaning-role assignment.

Table 5: Autoresearch with Gemma 4 26B-A4B-IT. Single run per condition. Baseline: pipeline from [3].

Game	Target	U	E	$\min_i R_i$	Keep rate
Cleanup	Baseline	-10.0	1.00 [†]	-1000	—
	Φ_U	0.87	-1.11	-371	6/19
	$\Phi_{\min}$	1.71	0.94	137	9/17
Gathering	Baseline	0.0	1.00 [†]	0	—
	Φ_U	2.44	0.98	580	4/11

[†]Equality is trivially 1.0 because all agents receive near-zero reward.

438 **Maximin optimization rescues efficiency.** Strikingly, under maximin optimization the researcher
439 achieves *higher* efficiency ($U=1.71$) than under direct efficiency optimization ($U=0.87$), alongside
440 strong equality ($E=0.94$) and positive worst-off welfare ($\min_i R_i=137$). This reversal—absent in
441 frontier models, where both objectives yield $U \approx 3.1-3.2$ —occurs because the maximin objective
442 forces the researcher toward coordination mechanisms (rotating cleaning duties, index-based apple

443 assignment) that simultaneously improve collective output. Under efficiency-only optimization, the
 444 researcher converges on a local optimum—static role assignment with 25% dedicated cleaners—that
 445 a 26B model can implement but that caps performance well below the frontier.

446 This suggests that *for models below a capability threshold, fairness objectives may serve as better*
 447 *optimization targets for overall social welfare than direct efficiency maximization.* The structured
 448 coordination enforced by the maximin objective provides a scaffold that compensates for the weaker
 449 model’s difficulty implementing complex strategies from strategic hints alone.

450 **Gathering: full recovery on a simpler game.** On Gathering ($N=4$), the researcher brings
 451 Gemma from $U=0.0$ to $U=2.44$ with $E=0.98$ and $\min_i R_i=580$, after 10 outer iterations (4 kept).
 452 These values nearly match frontier models (Gemini: $U=2.49$, Sonnet: $U=2.52$; Table 2). The
 453 researcher discovers the same Voronoi partitioning and respawn-aware camping strategies found
 454 in the main experiments, and increases inner-loop iterations to $K=5$ to allow sufficient refinement.
 455 The contrast with Cleanup—where Gemma peaks at $U=0.87/1.71$ versus frontier $U \approx 3.2$ —suggests
 456 that the researcher can fully compensate for model weakness on simpler coordination tasks (4 agents,
 457 symmetric costs) but not on harder provision dilemmas (10 agents, asymmetric costs).

458 D Researcher-Authored Pipeline Artifacts

459 The previous appendix shows the policies π_c^* that the inner loop *outputs*; this one shows the con-
 460 figuration $c = (p, \phi, \mathcal{H}, \iota, v)$ (Section 4, Table 1) that the researcher $\mathcal{R}$ *authored* to produce them.
 461 Excerpts are taken from the final commit on the dedicated git branch of the corresponding run; we
 462 reproduce the prose verbatim, with author commentary in [brackets] and elisions marked
 463 We organize by artifact type so each subsection makes a within-type contrast (e.g., Φ_U vs. $\Phi_{\min}$, or
 464 Cleanup vs. Gathering).

465 D.1 Helper library $\mathcal{H}$: coordination primitives

466 The researcher adds primitives that the policy LLM can call as black boxes, sparing it from re-
 467 implementing tricky logic on every iteration. Two patterns dominate: (i) *state inspection* (waste
 468 counts, fractions, beam-yield scoring) and (ii) *coordination primitives* (zone assignment, role rota-
 469 tion). We show one of each.

Listing 1: Cleanup, $\Phi_{\min}$ – band-based apple zoning helper added by the researcher in exp5. This is the primitive that lets the policy in Listing 6 keep collectors within their own row band, preventing all 7 gatherers from racing to the same apple.

```

470 1 def get_my_apples(env, agent_id):
471 2     """Alive apples in this agent's horizontal row band.
472 3     Divides the orchard into n_agents bands; falls back to all apples if empty."""
473 4     aid = int(agent_id)
474 5     n = int(env.n_agents)
475 6     band_h = env.height / n
476 7     band_lo, band_hi = aid * band_h, (aid + 1) * band_h
477 8
478 9     my_apples, all_apples = set(), set()
479 10    for i in range(env.n_apples):
480 11        if env.apple_alive[i]:
481 12            ar = int(env._apple_pos[i, 0]); ac = int(env._apple_pos[i, 1])
482 13            all_apples.add((ar, ac))
483 14            if band_lo <= ar < band_hi:
484 15                my_apples.add((ar, ac))
485 16    return my_apples if my_apples else all_apples
486

```

Listing 2: Gathering – BFS-Voronoi territory + respawn-aware waiting helpers added by the researcher in gather-exp3. These are the primitives that Listing 8 composes into a one-line policy: “go to my_zone_apples, otherwise nearest_respawning_apple.”

```

488 1 def voronoi_zones(env):
489 2     """Multi-source BFS Voronoi over walkable cells.
490 3     Returns (row, col) -> agent_id; ties broken by lower agent_id."""
491 4     queue, visited = deque(), {}
492 5     for a_id in range(env.n_agents):
493 6         if int(env.agent_timeout[a_id]) == 0:
494 7             ar = int(env.agent_pos[a_id][0]); ac = int(env.agent_pos[a_id][1])
495 8             queue.append((ar, ac, a_id))
496

```

```

497     visited[(ar, ac)] = a_id
498 while queue:
499     r, c, a_id = queue.popleft()
500     for dr, dc in [(-1,0),(1,0),(0,-1),(0,1)]:
501         nr, nc = r + dr, c + dc
502         if 0 <= nr < env.height and 0 <= nc < env.width \
503             and not env.walls[nr, nc] and (nr, nc) not in visited:
504             visited[(nr, nc)] = a_id
505             queue.append((nr, nc, a_id))
506     return visited
507
508 def nearest_respawning_apple(env, agent_id, zones=None, max_timer=10):
509     """Nearest dead apple in MY zone respawning within max_timer steps.
510     Returns (row, col) of best wait spot, or None."""
511     if zones is None: zones = voronoi_zones(env)
512     best_pos, best_t, best_d = None, max_timer + 1, float('inf')
513     ar = int(env.agent_pos[agent_id][0]); ac = int(env.agent_pos[agent_id][1])
514     for i in range(env.n_apples):
515         if not env.apple_alive[i]:
516             pos = (int(env._apple_pos[i][0]), int(env._apple_pos[i][1]))
517             if zones.get(pos) == agent_id:
518                 t = int(env.apple_timer[i])
519                 if t <= max_timer:
520                     d = abs(pos[0] - ar) + abs(pos[1] - ac)
521                     if t < best_t or (t == best_t and d < best_d):
522                         best_pos, best_t, best_d = pos, t, d
523     return best_pos

```

The choice of helper depends on the dilemma’s geometry: row-bands suffice when fairness is the main concern (Cleanup maximin) but full BFS-Voronoi is needed when wall-aware territory ownership matters (Gathering). The researcher does not fix this in advance — it picks the simpler primitive whenever it works.

D.2 System prompt p : from neutral framing to strategic briefing

The unmodified system prompt is a neutral API description: “Write a policy that maximizes per-agent reward.” Under maximin optimization, the researcher rewrites the prompt into a multi-section strategic briefing. The excerpt below shows the pieces it added to the Cleanup prompt in exp5; the full prompt grew from 165 to 325 lines.

Listing 3: Cleanup, $\Phi_{\min}$ – excerpts from the researcher-rewritten system prompt (exp5). The unmodified baseline contained only API documentation; everything below was added by $\mathcal{R}$.

```

534 ## CRITICAL OBJECTIVE: Rawlsian Fairness (Maximin)
535
536 Your goal is to maximize the minimum per-agent total return across all
537 agents. This is the "maximin" or Rawlsian welfare criterion.
538
539 This means: it is NOT enough for the *average* agent to do well. The
540 worst-off agent must do as well as possible. A policy where 8 agents
541 each earn +200 but 2 agents each earn -100 is TERRIBLE (maximin = -100).
542
543 Key implication: cleaning costs (-1 per CLEAN action) must be shared
544 equitably among ALL agents. If you assign fixed "cleaner" roles, those
545 agents will accumulate large negative rewards and destroy your maximin score.
546
547 ## Strategy for Maximin: ROLE ROTATION + APPLE ZONING
548
549 The optimal strategy for maximin involves:
550
551 1. Shared cleaning duty: ALL agents take turns cleaning using a rotation
552    schedule based on 'agent_id' and 'env.step_count'. For example:
553    'is_my_cleaning_turn = (agent_id + env.step_count // SHIFT) % env.n_agents < NUM_CLEANERS'
554    where SHIFT is ~50 steps and NUM_CLEANERS is 2-3.
555 2. When NOT your turn: collect apples from YOUR ZONE using
556    'get_my_apples(env, agent_id)' -- prevents all gatherers competing
557    for the same nearest apple.
558 3. NEVER use BEAM (action 6): it costs -1 and causes -50 to the target.
559 4. Keep waste density low: aim for ~2-3 active cleaners at any time.
560
561 [... API documentation, helper documentation, working example ...]
562
563 IMPORTANT:
564 - NEVER assign permanent cleaning roles to specific agents -- this kills maximin.
565 - Use env.step_count for time-based role rotation so all agents share cleaning duty.
566 - NEVER use BEAM (action 6) -- it destroys both agents' rewards.

```

Two things stand out. First, the researcher writes the formula it expects $\mathcal{M}$ to use almost verbatim — (agent_id + env._step_count // SHIFT) % env.n_agents < NUM_CLEANERS — and the policy in Listing 6 uses essentially this template. Second, the researcher *teaches by counter-example*: the explicit “8 agents earn +200, 2 earn -100, maximin = -100” makes the failure mode of Φ_U -style static roles (Listing 5) concretely visible to $\mathcal{M}$. This is the information-design move predicted by the mechanism-design framing in Section 3.4.

The Gathering prompt evolves along a different axis (no maximin, no rotation). Its researcher-added “Key Strategic Insights” section instead emphasizes (i) *self-play implies never beam* and (ii) *Voronoi partition each step*, exactly matching what the policy in Listing 8 implements.

D.3 Feedback ϕ : adaptive diagnostics

The baseline feedback function from [3] shows reward + four social metrics with definitions and stops there. The researcher turns it into a state machine: it inspects the latest metrics and conditionally injects different hints. Two different runs produced two qualitatively different diagnostics — the same metric ($\bar{r}_k$ or $\min_i R_i$) is repurposed to either *stabilize* a working policy or *redirect* a failing one.

Listing 4: Adaptive feedback diagnostics. *Top*: efficiency-optimized run (exp1) – a stability guard prevents iter-3 regression once U is high. *Bottom*: maximin-optimized run (exp5) – two fairness diagnostics that fire when the worst-off agent loses reward.

```

1 # --- Cleanup, Phi_U feedback (exp1, pipeline/feedback.py final) ---
2 last_eff = history[-1]["metrics"]["efficiency"]
3 if last_eff >= 2.5:
4     parts.append(
5         """CRITICAL -- DO NOT REGRESS*: The current policy achieves high "
6         "efficiency (>=2.5). You MUST output a policy that is nearly identical "
7         "to the current one. Copy the current policy code and make AT MOST "
8         "one small targeted improvement (e.g., adjust a single numeric "
9         "threshold). If you are not confident a change will help, output the "
10        "current policy UNCHANGED. A regression here means the run fails.")
11
12 # --- Cleanup, Phi_min feedback (exp5, pipeline/feedback.py final) ---
13 last_maximin = history[-1]["metrics"].get("maximin", 0)
14 last_avg = history[-1]["reward_avg"]
15 if last_maximin < 0:
16     parts.append(
17         f"""FAIRNESS ALERT*: maximin={last_maximin:.1f} is NEGATIVE. "
18         f"The worst-off agent lost reward overall while average was {last_avg:.1f}. "
19         f"This means cleaning duties are NOT shared equitably. "
20         f"Use time-based role rotation (env._step_count) so ALL agents share "
21         f"cleaning costs equally. NEVER assign permanent cleaner roles.")
22 elif last_maximin < last_avg * 0.5:
23     parts.append(
24         f"""FAIRNESS WARNING*: maximin={last_maximin:.1f} is much lower than "
25         f"average={last_avg:.1f}. The gap suggests unequal cleaning burden. "
26         f"Ensure ALL agents rotate through cleaning duty.")

```

The two diagnostics are written by independent researcher runs but converge on a common pattern: *thresholded interventions on a primary metric*. They are also a common cause of the low run-to-run variance reported in Finding 1: when a working policy lands in the high- U basin, the stability guard prevents the LLM from over-refining and tipping out of it (the $K=3$ regressions visible in Section 4’s “Common failure modes” (4)), and when an unfair policy lands in the negative-maximin region, the alert injects exactly the rotation hint that recovers it.

D.4 Iteration logic ι : per-condition hyperparameters

Although ι has the smallest source surface, the researcher’s edits to it explain a meaningful fraction of the variance reduction noted in Finding 1. Table 6 summarizes the values shipped in the final commit of each best-performing run.

The maximin Gemini run is the most informative case: the researcher discovered that with $K=3$ and $|S|=5$, identical configurations could yield $\min_i R_i=295$ on one set of seeds and $\min_i R_i=100$ on another (this is the “variance exposure” failure mode of Section 4’s common failures). It then walked $|S|$ from 5 $\rightarrow$ 8 $\rightarrow$ 12 over three outer iterations until the seed-averaged signal was stable enough that the policy LLM stopped chasing noise. The policy in Listing 6 is sampled at $|S|=12$.

Table 6: Iteration-logic (ι) values in the final commit of selected runs. K = inner-loop iterations, $|S|$ = evaluation seeds, “thinking” = extended-thinking token budget passed to $\mathcal{M}$. Baseline ([3]) is $K=3$, $|S|=5$, thinking 16k.

Run	Game / objective	K	$ S $	thinking	Researcher’s stated rationale
exp1 (Gemini)	Cleanup, Φ_U	3	5	16k	default; stability comes from feedback hint
exp4 (Sonnet)	Cleanup, Φ_U	3	5	32k	“Sonnet was over-refining at 16k thinking”
exp5 (Gemini)	Cleanup, $\Phi_{\min}$	2	12	16k	“ $K=2$ avoids iter-3 regression; 12 seeds for variance control”
exp7 (Sonnet)	Cleanup, $\Phi_{\min}$	3	5	10k	“reduced thinking from 16k – Sonnet was generating runaway code at 16k”
gather-exp3 (Gemini)	Gathering, Φ_U	3	5	16k	default; Gathering converges in $K \leq 3$

D.5 Cross-artifact observations

- *Helpers do the algebra; prompts pick the strategy class.* The researcher uses helpers (1, 2) to put non-trivial primitives at $\mathcal{M}$ ’s fingertips, and the prompt (3) to tell $\mathcal{M}$ *which* primitive to compose. Neither is sufficient alone — prompts without supporting helpers produced policies that “mention rotation” but failed to implement it; helpers without prompt updates often went unused.
- *Feedback is the only adaptive component.* p , $\mathcal{H}$, and ι are static within a run; ϕ is the only place where the researcher gets to change what $\mathcal{M}$ sees *during* the inner loop, and it uses thresholded diagnostics (Listing 4) to do so. This is what mostly drives the run-to-run variance reduction.
- *The same artifact has different content under each objective.* The Cleanup prompt under Φ_U omits “rotation” and “maximin” entirely; the same prompt under $\Phi_{\min}$ devotes its entire opening section to it. The configuration c is genuinely a function of the welfare objective Φ , not a fixed pipeline parameterized by it — consistent with the information-design framing of Section 3.4.
- *The researcher’s edits are short and targeted.* The full diff between the baseline pipeline and the best maximin Gemini configuration totals ~ 300 added lines spread over four files; ~ 80 of them appear in the listings above. The remaining content is API documentation and worked examples, included for completeness but not strategically novel.

E Selected Generated Policies

The code excerpts below are verbatim outputs of the policy synthesizer $\mathcal{M}$, taken from the final inner-loop iteration of the best-performing run in each condition. To fit the page, we elide repeated boilerplate (BFS scaffolding, fallback branches) with `...` and add commentary in [brackets]; nothing else has been edited. The listings illustrate the qualitative differences highlighted in Findings 2–4 of Section 4: *static* vs. *rotating* cleaning roles in Cleanup, and the *Voronoi* + *respawn-timer* structure in Gathering. All four policies were independently generated by the LLM under different seeds, models, and outer-loop runs.

E.1 Cleanup, Φ_U (Gemini): static interleaved roles + dynamic threshold

Best run: $U=3.25$, $E=0.61$, $\min_i R_i = -182$. The synthesizer assigns *permanent* cleaning roles via `agent_id mod n <  $\tau$` , where the threshold τ depends smoothly on the current waste fraction. Higher pollution recruits more cleaners, but the *identity* of each cleaner is fixed — agents $0-\tau-1$ always pay the cost. This drives U to its ceiling but leaves equality at 0.61 and pushes the worst-off agent to $\min_i R_i = -182$.

Listing 5: Cleanup, Φ_U – Gemini 3.1 Pro. Static role assignment with waste-tiered threshold (verbatim, abridged).

```

def policy(env, agent_id) -> int:
    if int(env.agent_timeout[agent_id]) > 0:
        return 7 # STAND while removed
    waste_frac = waste_fraction(env)
    n_agents = env.n_agents
    # [Smooth dynamic threshold: more pollution -> more cleaners]
    if waste_frac > 0.35: threshold_cleaners = n_agents
    elif waste_frac > 0.25: threshold_cleaners = int(n_agents * 0.7)
    elif waste_frac > 0.15: threshold_cleaners = int(n_agents * 0.5)
    elif waste_frac > 0.05: threshold_cleaners = int(n_agents * 0.3)

```

```

669 13     else:                                     threshold_cleaners = max(1, int(n_agents * 0.2))
670 14
671 15     # [STATIC role: cleaners are ALWAYS the lowest-id agents -- no rotation]
672 16     is_cleaner = (agent_id % n_agents) < threshold_cleaners
673 17
674 18     if is_cleaner:
675 19         best_orient, n_waste = find_best_clean_orientation(env, agent_id)
676 20         if n_waste >= 1:
677 21             cur_orient = int(env.agent_orient[agent_id])
678 22             if best_orient == cur_orient: return 8 # CLEAN
679 23             if (cur_orient + 1) % 4 == best_orient: return 5 # ROTATE_RIGHT
680 24             elif (cur_orient - 1) % 4 == best_orient: return 4 # ROTATE_LEFT
681 25             else: return 4
682 26         # [Competitor-aware BFS toward waste in own row band]
683 27         c_idx = agent_id % n_agents
684 28         target_r = int(((c_idx + 0.5) / threshold_cleaners) * env.height)
685 29         # ... pick waste cell minimizing dist + |row - target_r| + 50 *
686 30         # closer_cleaners
687 31         # ... bfs_toward and direction_to_action
688 32     else:
689 33         # GATHERER: row-banded sweep, competitor-aware
690 34         assigned_row = ((g_idx + 0.5) / n_gatherers) * env.height
691 35         # ... pick alive apple minimizing dist + |row - assigned_row| + 50 *
692 36         # closer_gatherers
693 37     ...

```

695 E.2 Cleanup, $\Phi_{\min}$ (Gemini): time-rotated, geographically interleaved roles

696 Best run: $U=3.19$, $E=0.98$, $\min_i R_i=296$. Switching the outer objective from U to $\min_i R_i$ flips
697 the role-assignment scheme entirely: `role_idx` now depends on `agent_id + step/50`, so every
698 agent rotates through cleaner and gatherer duty in a 50-step cycle. The cleaner indices $\{1, 4, 8\}$ are
699 spatially *interleaved* along the river (rather than contiguous) so that whoever is on duty has a short
700 walk to their assigned river slice. This single change accounts for the $E=0.55 \rightarrow 0.98$ jump and the
701 $\min_i R_i=-182 \rightarrow 296$ swing reported in Table 2, with negligible efficiency loss.

Listing 6: Cleanup, $\Phi_{\min}$ – Gemini 3.1 Pro. Time-rotated roles with interleaved geography (verbatim, abridged).

```

702 1 def policy(env, agent_id) -> int:
703 2     if int(env.agent_timeout[agent_id]) > 0:
704 3         return 7 # STAND while removed
705 4
706 5     aid = int(agent_id)
707 6     step = int(env.step_count)
708 7     n = int(env.n_agents)
709 8
710 9     # [TIME ROTATION: every 50 steps each agent's role index advances]
711 10    if n == 10:
712 11        role_idx = (aid + step // 50) % 10
713 12        # [Cleaner indices {1,4,8} are SPATIALLY INTERLEAVED, not contiguous,
714 13        # so each on-duty cleaner has a short walk to its river slice]
715 14        if role_idx == 1: role_type, zone_idx = 'C', 0
716 15        elif role_idx == 4: role_type, zone_idx = 'C', 1
717 16        elif role_idx == 8: role_type, zone_idx = 'C', 2
718 17        else:
719 18            role_type = 'G'
720 19            g_map = {0:0, 2:1, 3:2, 5:3, 6:4, 7:5, 9:6}
721 20            zone_idx = g_map[role_idx]
722 21            num_cleaners, num_gatherers = 3, 7
723 22
724 23    if role_type == 'C':
725 24        # [Each on-duty cleaner owns one horizontal slice of the river]
726 25        row_min = int(zone_idx * env.height / num_cleaners)
727 26        row_max = int((zone_idx + 1) * env.height / num_cleaners) - 1
728 27
729 28        # [Score every (row, col, orientation) in the slice by beam yield;
730 29        # fire if facing >=2 waste, else move to a >=2-yield position]
731 30        best_positions, good_positions = set(), set()
732 31        for r in range(row_min, row_max + 1):
733 32            for c in range(10):
734 33                if env.walls[r, c] or env.waste[r, c]: continue
735 34                y_max = max(get_clean_yield(env, r, c, o) for o in range(4))
736 35                if y_max >= 2: best_positions.add((r, c))
737 36                if y_max >= 1: good_positions.add((r, c))
738 37            # ... rotate / CLEAN / bfs_to_target_set as appropriate
739 38        else:
740 39            # GATHERER: collect apples ONLY inside the rotating row band

```

```

742 40 # row_min/row_max for num_gatherers slices ...
743 41 ...

```

745 E.3 Cleanup, $\Phi_{\min}$ (Sonnet): independent rediscovery of duty rotation

746 Best run: $U=2.93$, $E=0.83$, $\min_i R_i=154$. Run with a different policy LLM (Sonnet 4.6) and on
747 a different outer-loop trajectory, the synthesizer arrives at the *same* structural insight as the Gemini
748 maximin run: a phase counter ($\text{agent_id} + \text{step}/50$) mod n rotates which 2 of the 10 agents clean
749 at any given time, and a *separate* 200-step zone counter rotates which 5-row band each collector
750 sweeps. The two rotation periods (50 and 200 steps) ensure every agent visits every cleaning slot
751 *and* every apple zone within one episode — a structural fairness invariant the researcher discovered
752 without any explicit instruction to “rotate.”

Listing 7: Cleanup, $\Phi_{\min}$ – Sonnet 4.6. Phase-rotated cleaning + zone-rotated collection (verbatim, abridged).

```

753 1 def policy(env, agent_id) -> int:
754 2     if int(env.agent_timeout[agent_id]) > 0:
755 3         return 7 # STAND while removed
756 4
757 5     n = env.n_agents
758 6     step = env._step_count
759 7     wf = waste_fraction(env)
760 8
761 9     # --- ROLE ASSIGNMENT: fair 2-cleaner rotation ---
762 10    # [50-step phases x 20 phases / episode -> each agent cleans 4 phases (~20%)]
763 11    phase = step // 50
764 12    cleaner_rank = (agent_id + phase) % n # [cycles 0..9 fairly]
765 13    is_cleaner = cleaner_rank < 2 # [exactly 2 cleaners per phase]
766 14
767 15    # [Emergency override: all agents clean when waste crosses the spawn cliff]
768 16    in_emergency = wf >= 0.40
769 17    if in_emergency: is_cleaner = True
770 18
771 19    # [River split: cleaner_slot 0 -> top half, 1 -> bottom half]
772 20    cleaner_slot = (agent_id % 2) if in_emergency else cleaner_rank
773 21
774 22    if is_cleaner and wf > 0.04:
775 23        # [Scan 4 orientations from current position; if best beam shot
776 24        # covers >=3 waste cells, fire; else step to a strictly better neighbor]
777 25        # ... (best_o, best_cnt) = max over 4 directions
778 26        # ... if best_cnt >= 3: rotate to best_o, then CLEAN
779 27        # ... else: scan 4 adjacent cells x 4 orientations for higher yield
780 28        # ... if no waste in beam range: bfs to waste_set in OWN river half,
781 29        # fall back to whole river
782 30
783 31    # --- COLLECTOR LOGIC: 5-zone rotating apple collection ---
784 32    # [5 zones x 200 steps = 1000 steps -> each agent visits every zone once;
785 33    # 2 agents share each zone at any time, reducing competition 10 -> 2]
786 34    bounds = zone_boundaries(env, 5)
787 35    zone_phase = step // 200
788 36    zone = (agent_id + zone_phase) % 5
789 37    z_start, z_end = bounds[zone], bounds[zone + 1]
790 38
791 39    zone_apples = {(int(env._apple_pos[i][0]), int(env._apple_pos[i][1]))
792 40                  for i in range(env.n_apples) if env.apple_alive[i]
793 41                  and z_start <= int(env._apple_pos[i][0]) < z_end}
794 42    # ... bfs_to_target_set(zone_apples), else bfs_nearest_apple as fallback
795 43    return 7
796

```

798 E.4 Gathering (Gemini): wall-aware Voronoi + spatiotemporal targeting

799 Best run: $U=2.47$, $E=0.98$, $\min_i R_i=580$. In Gathering, where costs are symmetric, no role dif-
800 ferentiation is needed: the entire optimization is spatial and temporal. The synthesizer (i) parti-
801 tions cells into BFS-Voronoi territories owned by individual agents, (ii) runs a single centralized
802 BFS from its own position to score every reachable cell with exact wall-aware distances, and
803 (iii) for both alive and respawning apples in its territory, computes the *earliest collection time*
804 $\max(\text{walk_dist}, \text{respawn_timer})$ and targets the minimum. When the agent must wait on a
805 dead apple, it steps onto a *non-spawn* adjacent cell rather than blocking a respawn point. This single
806 sort key — $\max(\text{distance}, \text{timer})$ — subsumes both “go to nearest apple” and “camp respawn” as
807 special cases.

Listing 8: Gathering – Gemini 3.1 Pro. Voronoi territories + spatiotemporal priority targeting (verbatim, abridged).

```

808
809 1 def policy(env, agent_id) -> int:
810 2     if int(env.agent_timeout[agent_id]) > 0:
811 3         return 7
812 4
813 5     r, c = int(env.agent_pos[agent_id][0]), int(env.agent_pos[agent_id][1])
814 6     orient = int(env.agent_orient[agent_id])
815 7
816 8     # [Wall-aware Voronoi over the gridworld; spawn_points = all apple cells]
817 9     zones = voronoi_zones(env)
818 10    spawn_points = {(int(p[0]), int(p[1])) for p in env._apple_pos}
819 11
820 12    # [Single centralized BFS: exact distance + first move to every reachable cell]
821 13    distances, first_moves = {}, {}
822 14    q = deque([(r, c, 0, None)])
823 15    visited = {(r, c)}
824 16    while q:
825 17        cr, cc, d, fm = q.popleft()
826 18        distances[(cr, cc)] = d
827 19        first_moves[(cr, cc)] = fm
828 20        for dr, dc in [(-1,0),(1,0),(0,-1),(0,1)]:
829 21            nr, nc = cr + dr, cc + dc
830 22            if 0 <= nr < env.height and 0 <= nc < env.width and not env.walls[nr,
831 23                nc]:
832 24                if (nr, nc) not in visited:
833 25                    visited.add((nr, nc))
834 26                    q.append((nr, nc, d + 1, fm if fm is not None else (dr, dc)))
835 27
836 28    # [Both alive AND respawning apples in MY zone, with their respawn timer]
837 29    my_zone_spawns = []
838 30    for i in range(env.n_apples):
839 31        pr, pc = int(env._apple_pos[i][0]), int(env._apple_pos[i][1])
840 32        if zones.get((pr, pc)) == agent_id and (pr, pc) in distances:
841 33            my_zone_spawns.append((pr, pc, int(env.apple_timer[i]), distances[(pr,
842 34                pc)]))
843 35
844 36    # [SPATIOTEMPORAL PRIORITY: earliest collection time first.
845 37    # score = max(walk_distance, respawn_timer); ties broken by sooner timer]
846 38    my_zone_spawns.sort(key=lambda x: (max(x[3], x[2]), x[2], x[3]))
847 39
848 40    for pr, pc, timer, dist in my_zone_spawns:
849 41        if timer == 0:
850 42            if dist == 0: return 7
851 43            fm = first_moves[(pr, pc)]
852 44            return direction_to_action(fm[0], fm[1], orient)
853 45        else:
854 46            # [Dead apple: camp on safe adjacent cell]
855 47            # ... pick nearest neighbor (nr,nc) that is NOT in spawn_points,
856 48            # ... navigate there and STAND while waiting for respawn
857 49            # [Global poaching fallback if zone is empty: nearest alive apple anywhere]
858 50            ...

```

860 E.5 Cross-condition observations

861 Several patterns are visible across these four listings (and confirmed by inspecting the remaining 8
862 runs not shown):

- 863 • *Role assignment encodes the welfare objective.* Static $\text{agent_id} < \tau$ (Listing 5) maximizes
864 U but harms $\min_i R_i$. Time-rotated $(\text{agent_id} + \text{step} // T) \% n$ (Listings 6, 7) maximizes
865 $\min_i R_i$ at $\leq 1\%$ efficiency cost (Gemini).
- 866 • *Convergent rediscovery across LLMs.* Gemini and Sonnet, on independent runs, both arrive at
867 the $(\text{id} + \text{phase}) \% n$ duty-rotation idiom with similar phase lengths ($T \approx 50$ steps) under
868 maximin — and both omit it under efficiency.
- 869 • *Game structure dictates strategy class.* Cleanup policies are dominated by role-assignment logic
870 (cleaner vs. gatherer); Gathering policies are dominated by territory partitioning and respawn-
871 timer reasoning, with no role differentiation at all. The researcher does not need to be told which
872 class of strategy to write.
- 873 • *Earliest-collection-time as a unifying score.* The Gathering policy’s $\max(\text{walk}, \text{timer})$ key (List-
874 ing 8) is structurally identical across the 4 Gathering runs (both Gemini and Sonnet), differing
875 only in how the Voronoi diagram is computed — Manhattan approximation, fully BFS-based, or
876 cached helper.